Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/07/2017
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	The Role of Vacuole Membrane Fusion in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
PI Email:	mrojasp@ncsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919-513-8044
Organization Name:	North Carolina State University		
PI Address 1:	Plant Biology		
PI Address 2:	2115 Gardner Hall -Box 7612		
PI Web Page:			
City:	Raleigh	**State:**	NC
Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	07/01/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/1/2018 per NSSC information (Ed., 8/24/17) NOTE: Extended to 7/1/2017 per NSSC information (Ed., 4/5/16)		
Key Personnel Changes/Previous PI:	June 2017 report: Dr. Jiameng Zheng left the lab August 1st, 2016 ; Dr. Carla Brillada was hired on February 1st, 2017.		
COI Name (Institution):			
Grant/Contract No.:	NNX13AM49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The long-term goal of this research is to identify and characterize mechanisms of plant vacuole biogenesis that are critical for gravity perception. A role of the vacuole in controlling gravity perception is well accepted, but the molecular mechanisms by which the vacuole interacts with amyloplasts or regulates their sedimentation during gravitropism are yet to be identified. We recently uncovered a novel vacuole phenotype in agravitropic mutant alleles of VTI11 in Arabidopsis. We predict that this vacuole phenotype of vti11 mutants has negative effects on the organization of actin cables or the cortical endoplasmic reticulum, and therefore prevents amyloplast sedimentation. In addition, we have uncovered a novel regulation of vacuolar membrane fusion by phosphoinositides. Based on our results, we developed the hypothesis that mechanisms of vacuole remodeling that are mediated by phosphoinositide signaling are critical for the perception or transduction of the gravitropic signal in plants. A unique assay for vacuole fusion that we developed will be used to determine the role of specific cellular components in membrane fusion and gravity perception. The specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. Overall, this research will improve our understanding of vacuole biogenesis, the interactions between the vacuole and other cellular components involved in gravity perception, and the role of phosphoinosidides in gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. In addition, the function of the vacuole in establishing the morphology and dynamics of other organelles will be characterized. These results will enhance the fundamental understanding of how plant cells organize into a defined set of independent organelles.
Task Progress:	We finalized a revision for Alvarez et al. (2016) manuscript, and this was accepted in October 2017 in the Journal of Experimental Botany. This publication described the consequences of the fragmented vacuole of zig1 for amyloplast movement, cytoskeleton organization, and ER (endoplasmic reticulum) in membrane dynamics. A major conclusion of this work was that amyloplasts in zig1 are extremely static and even resist sedimentation after a force of 10g. We hypothesized that this characteristic of zig1 mutants was due to the physical tethering of amyloplasts to vacuoles, which we proposed is indirect and involves physical associations between both amyloplasts and vacuoles with the ER. In fact, we showed that neither actin nor microtubule cytoskeleton were involved in this tethering. This last year we have focused on testing this hypothesis using forward and reverse genetic approaches. In addition, we have continued to characterize the interaction between VTI11 SNARE and the HOPS complex and a manuscript that describes this work has been finalized.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Articles in Peer-reviewed Journals	Alvarez AA, Han SW, Toyota M, Brillada C, Zheng J, Gilroy S, Rojas-Pierce M. "Wortmannin-induced vacuole fusion enhances amyloplast dynamics in Arabidopsis zigzag1 hypocotyls." J Exp Bot. 2016 Dec;67(22):6459-6472. https://doi.org/10.1093/jxb/erw418 ; PubMed PMID: 27816929; PubMed Central PMCID: PMC5181587 , Dec-2016